CASE STUDY 6

Comparative Analysis of Classification Algorithms for Bank Marketing Prediction

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Subject	Fundamentals of Data Science (22CSC59)
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Dataset	Bank Marketing — bank-full.csv

A data science study comparing four classifiers for predicting term-deposit subscription from historical bank marketing records.

1. Problem Statement and Methodology

A Portuguese bank conducts telephone marketing campaigns to promote term deposits. The objective is to develop and compare classification models that predict whether a customer subscribes, then recommend a model based on analytical evidence and campaign needs.

Dataset and target. The supplied bank-full.csv file contains 45,211 records and 17 columns, including 16 customer/campaign attributes and target y. The file is semicolon-delimited. The target has 39,922 'no' outcomes (88.30%) and 5,289 'yes' outcomes (11.70%). No exact duplicate rows were detected. Several categorical fields use the literal category 'unknown'; these are retained as a category rather than silently treated as missing.

Preprocessing. The target was mapped to 0/1 (no/yes). The data were split into 80% training and 20% test sets using stratification and random_state=42. Numeric predictors were median-imputed and standardized; categorical predictors were imputed with the most frequent category and one-hot encoded. These transformations were fitted within each model pipeline.

Important predictor decision. The variable *duration* was excluded from all primary models because call duration is not known before the call takes place. Including it would not match realistic pre-contact targeting.

2. Exploratory Data Analysis

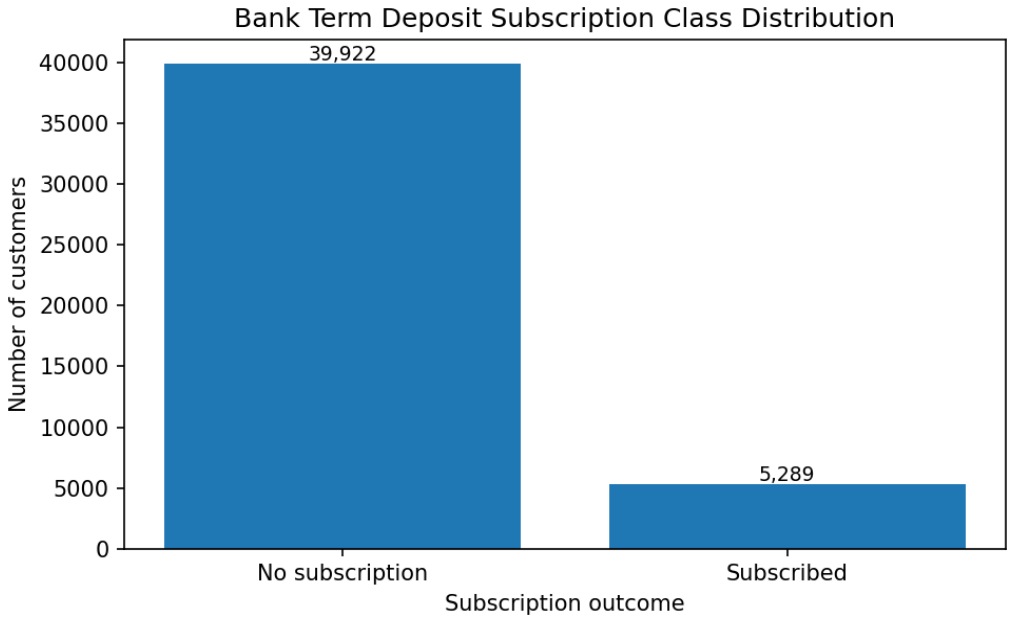


Figure 1. About 88.3% did not subscribe and 11.7% subscribed. A model can obtain high accuracy by mostly predicting 'no' while still missing many potential subscribers.

2. Exploratory Findings

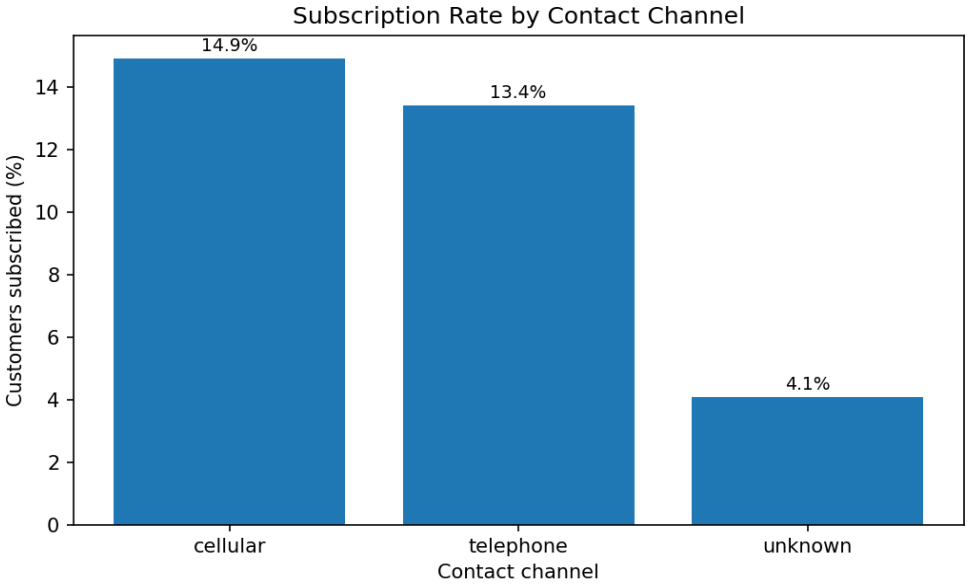


Figure 2. Historical subscription rates differ by contact channel. This is observational and does not establish that the channel itself causes subscription.

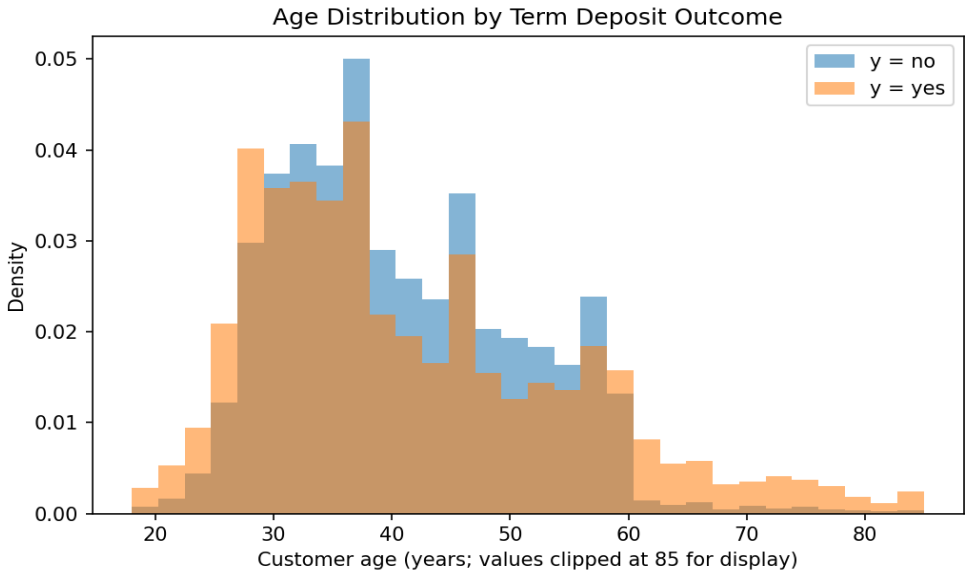


Figure 3. Age distributions overlap between outcomes; age alone does not separate subscribers from non-subscribers. Values above age 85 are clipped only for display.

Three evidence-based observations

1. The target is imbalanced: 11.70% subscribed versus 88.30% not subscribed. Evaluation should emphasize positive-class precision, recall and F1 as well as accuracy.

2. The observed subscription percentage varies by contact channel, suggesting channel and campaign attributes are worth examining when planning outreach.

3. Median age was 38 for subscribers and 39 for non-subscribers. This is descriptive, not causal.

3. Classification Models and Evaluation

Four Scikit-learn classifiers were trained on the same split: Logistic Regression, Decision Tree (min_samples_leaf=10), KNN (7 neighbours, distance weighting), and Random Forest (160 trees). Each model generated test predictions and probability scores.

Table 1. Held-out test-set performance

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Decision Tree	0.887	0.537	0.268	0.358	0.719
Random Forest	0.896	0.668	0.218	0.329	0.803
KNN	0.889	0.576	0.198	0.294	0.714
Logistic Regression	0.893	0.663	0.179	0.281	0.772

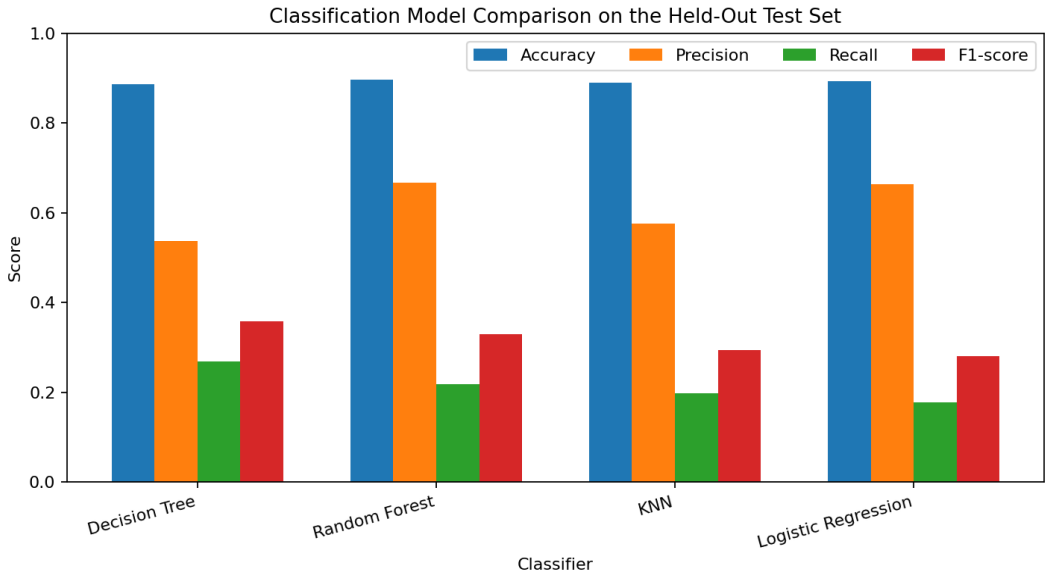
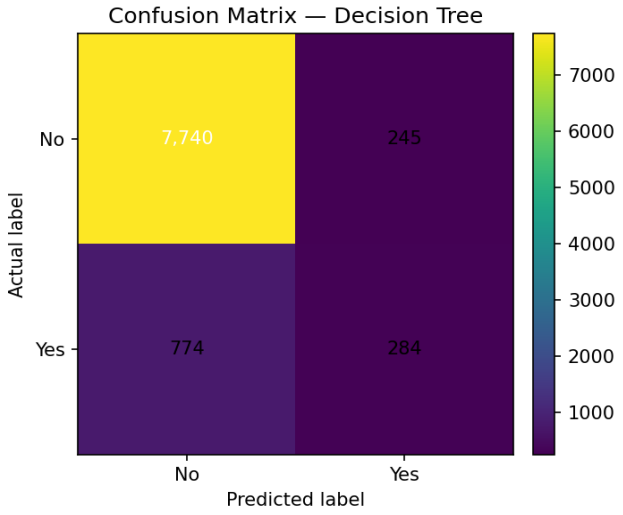


Figure 4. Decision Tree achieved the highest F1-score and recall at the tested default threshold; Random Forest achieved the highest ROC-AUC. No classifier dominates every metric.

4. Error Analysis, Recommendation and Limitations

Confusion matrix — Decision Tree. The model correctly classified 7,740 non-subscribers and 284 subscribers, while producing 245 false positives and 774 false negatives. False positives may lead to calls to customers who do not subscribe; false negatives may mean missing customers who would subscribe.



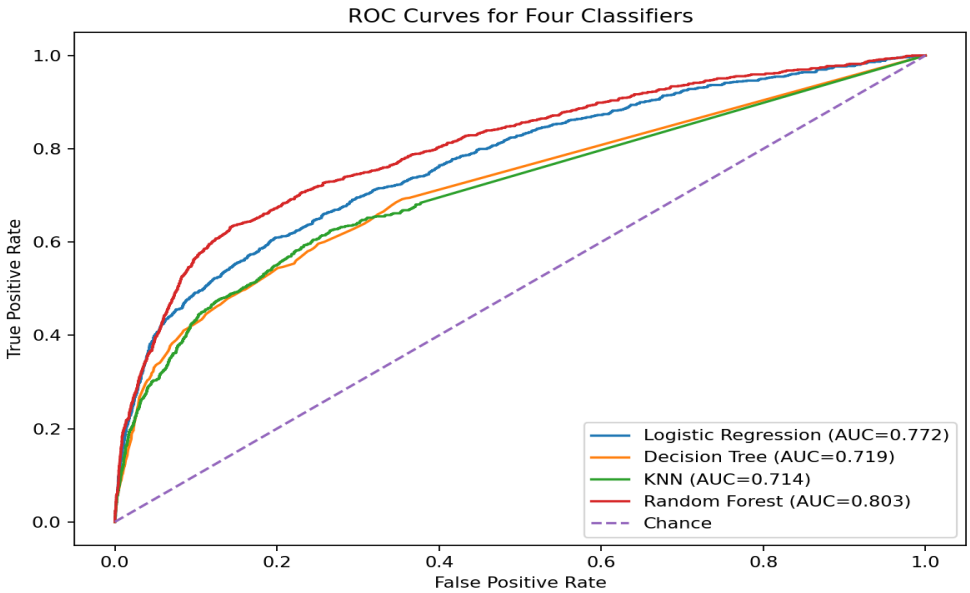


Figure 5. Random Forest has the highest ROC-AUC (0.803), indicating the strongest ranking performance in this comparison. ROC-AUC does not choose the operational threshold.

Recommendation. If the immediate objective is to balance precision and recall at the tested threshold, Decision Tree is the leading baseline (F1=0.358, recall=0.268). Random Forest should also be considered because it has the highest ROC-AUC (0.803). Validate threshold-tuned candidates against campaign costs before deployment.

Business recommendations

1. Pilot score-based targeting on a limited campaign, compare subscription rate and cost per subscription against the current approach, and use a control group to estimate incremental lift.

2. Tune the probability threshold according to the cost of unnecessary calls and the value of finding additional subscribers; monitor precision, recall, F1 and ROC-AUC over time.

Limitations

1. Historical data may not represent future customer behaviour, campaign policy or economic conditions; performance can drift.

2. The baseline comparison uses one holdout split, default thresholds and limited hyperparameter tuning. Class imbalance lowers recall, so cross-validation, class weighting, threshold tuning and calibration may change the ranking.

Conclusion. Decision Tree is the strongest baseline by F1/recall, while Random Forest ranks customers best by ROC-AUC. Recall at default thresholds is modest; the bank should validate and threshold-tune a candidate model against campaign costs before operational use.